

Threshold Detection - Medium Problem

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You are designing a system to check continuity on an e-match (electronic match) placed in a rocket engine to ignite the fuel. So far, you have designed a circuit that outputs 12V if there is no continuity, and outputs 1.5V if there is continuity. You are using a raspberry pi to measure if there is or is not continuity, and you do not want to have to write code to read an analog voltage and check if it is below a threshold. It would be much simpler if the circuit just output 0V if there is not continuity, and 5V if there is continuity.

Design a circuit that takes a signal that can either be 12V or 1.5V as the input, and outputs 0V if the input signal is 12V and outputs 5V if the input signal is 1.5V. The intermediate behavior of the circuit at other voltages is unimportant, and is left up to you.